The Extension Bonus

Do big extenders earn more called strikes than extension alone should give them?

Trent Reimers | MLB 2021-2026 | October 29, 2026

The question. Pitchers who release the ball closer to the plate seem to steal called strikes. Statcast already credits extension with a velocity boost: the ball covers less distance, so it plays like a faster pitch. Do big extenders get more called strikes than that boost alone would predict?

The short answer. Partly. On four-seamers, each foot of extension adds called strikes, while real velocity takes them away. So extension is doing something that velocity isn't, and it shows up as hitters taking more pitches in the zone. But the biggest extenders don't get anything on top of the steady per-foot gain, whether extension is measured in raw feet or against the pitcher's height.

Key findings

1. **Extension doesn't act like velocity.** One foot of extension plays like about 1.72 mph of extra velocity. But on four-seamers, real velocity *costs* called strikes (−0.43 per 100 pitches per mph), so velocity alone predicts a foot of extension should cost 0.73. Instead it adds +0.76. That is a gap of +1.49 called strikes per 100 four-seamers per foot ($p < 0.0001$).
2. **The extra strikes come from takes in the zone.** With height held constant, each foot of extension raises the four-seam in-zone take rate by 1.4 points, while each mph of velocity lowers it by 0.6. About a third of the gain survives once pitch location is accounted for (+0.35 called strikes above expected per foot).
3. **There is no big-extender bonus.** Pitchers past 7 ft (about the top 10%) sit on or slightly below the straight per-foot line. A curved fit flattens past 7 ft rather than bending up.
4. **Height doesn't change the story.** Height explains only 14% of who extends far. Holding height constant, the four-seam gain holds (+1.09 per foot). Pitchers who extend far *for their size* get no extra boost, and extension as a percentage of height adds nothing beyond raw extension.
5. **It's a four-seam and changeup effect, not a pitcher effect.** More extension also helps changeups and splitters, but it costs called strikes on cutters and breaking balls. Across a pitcher's whole arsenal, the net gain is about zero.

Data: 4.2 million MLB regular-season pitches, 2021-2026, grouped into 9,001 pitcher-season-pitch type groups (150+ pitches each) from 1,246 pitchers. Heights are listed heights from the MLB StatsAPI. Data and methods are in the appendix.

PART 1

What extension "should" be worth

Statcast turns extension into perceived velocity: on four-seamers, each extra foot adds 1.72 mph of effective speed. If extension mattered only through that, a foot of extension should move called strikes as much as 1.72 mph of real velocity does. That is the benchmark.

All models below compare pitchers who throw the same pitch type at the same velocity, IVB, HB, release height, release side and arm angle, in the same season. Two outcomes are used. **CS%** is called strikes per 100 pitches. **CSOE** is called strikes above what that pitch's location and count predict, which takes out where the pitcher threw it and leaves how hitters and umpires reacted.

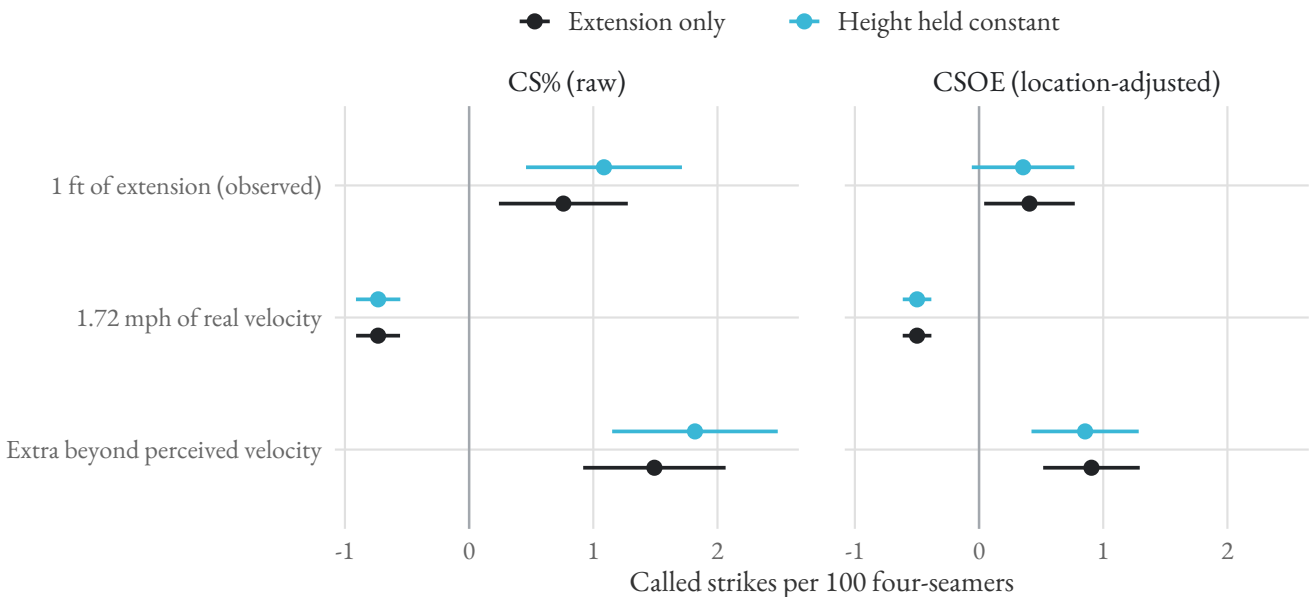


Figure 1: Four-seamers: called strikes per 100 pitches from one extra foot of extension, against what its perceived-velocity boost predicts (1.72 mph of real velocity). Bars are 95% intervals, clustered by pitcher.

Real velocity and extension point in opposite directions. Harder four-seamers get *fewer* called strikes (−0.43 per 100 per mph), because hitters swing at them more. Extension gets *more*. So the benchmark predicts a loss of 0.73 per foot, and the data show a gain of +0.76. The location-adjusted version tells the same story at about half the size (+0.41 observed against −0.50 predicted).

> **Takeaway.** Extension isn't just free velocity. If it were, longer extension would cost four-seam called strikes the way velocity does. It adds them instead: about 1.5 per 100 four-seamers per foot beyond the benchmark.

PART 2

Where the extra strikes come from

A called strike needs two things: a pitch in or near the zone, and a hitter who doesn't swing. The four-seam called strike gain can be split into those parts. Height is held constant here.

- **Takes in the zone carry it.** Extension raises the in-zone take rate by 1.4 points per foot. Real velocity at the same perceived-velo dose lowers it by 1.0. Hitters let strikes go by more often against long extenders.

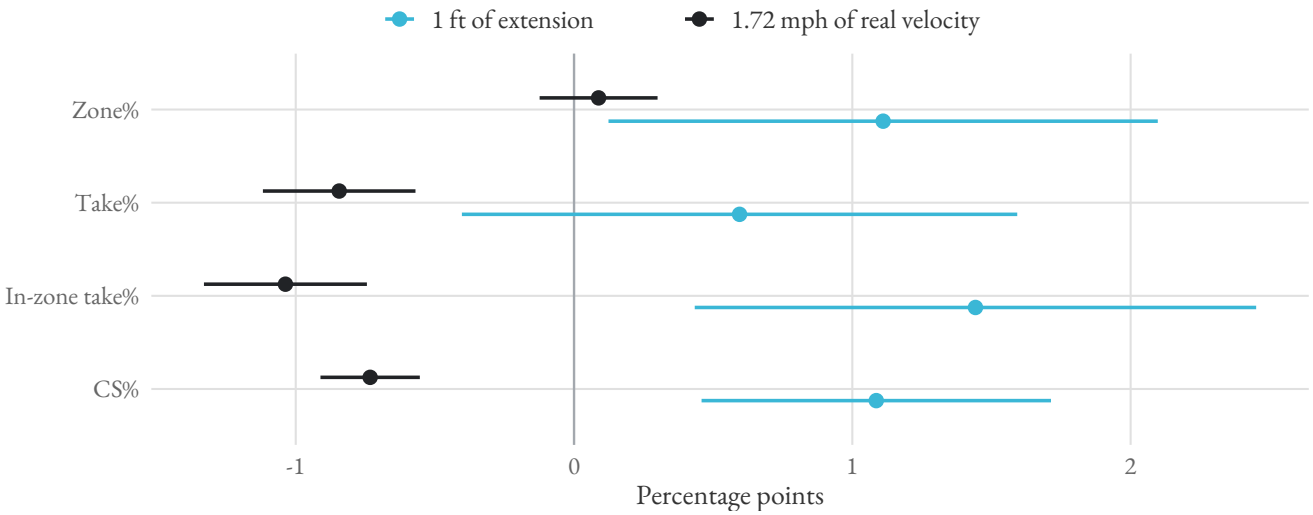


Figure 2: Four-seamers, height held constant: change in each rate (percentage points) from one foot of extension and from 1.72 mph of real velocity. Bars are 95% intervals.

- **Zone rate helps a little.** Long extenders throw slightly more four-seamers in the zone (+1.1 points per foot, p 0.049 against the benchmark). This is the part the location-adjusted CSOE removes, which is why CSOE runs a third to a half of raw CS%.

> **Takeaway.** The gain is mostly hitters taking strikes, which fits a deception or timing story more than a command one. Some of it is long extenders simply landing more four-seamers in the zone.

PART 3

Do the biggest extenders beat the line?

Part 1 fits one slope per foot. The question here is whether the biggest extenders sit above that line. The figure shows called strikes by extension bucket, with each pitcher's other traits taken out, alongside the straight line and a flexible curve.

- **The line holds through about 7 ft.** Up to there, each bucket sits close to the straight fit.
- **Past 7 ft it flattens.** For CSOE the curve levels off after 7 ft instead of rising (curvature test p 0.036; for raw CS% p 0.48). An extra slope for pitchers above 7 ft is about zero (−0.24 per foot for CS%, p 0.87).
- **The top bucket is small.** Above 7.5 ft there are only about 20 pitcher-seasons. On raw CS% that bucket sits above the line, but its interval easily covers the line, and the 7.0-7.25 ft bucket next to it sits below. On CSOE it lands on the line.

> **Takeaway.** Extension pays at a steady rate per foot, and returns fade at the top. The biggest extenders get the per-foot gain and nothing more.

PART 4

Extension relative to height

A 6'7" pitcher reaching 7 ft is doing something different from a 6'0" pitcher reaching 7 ft. If the edge is deception, it

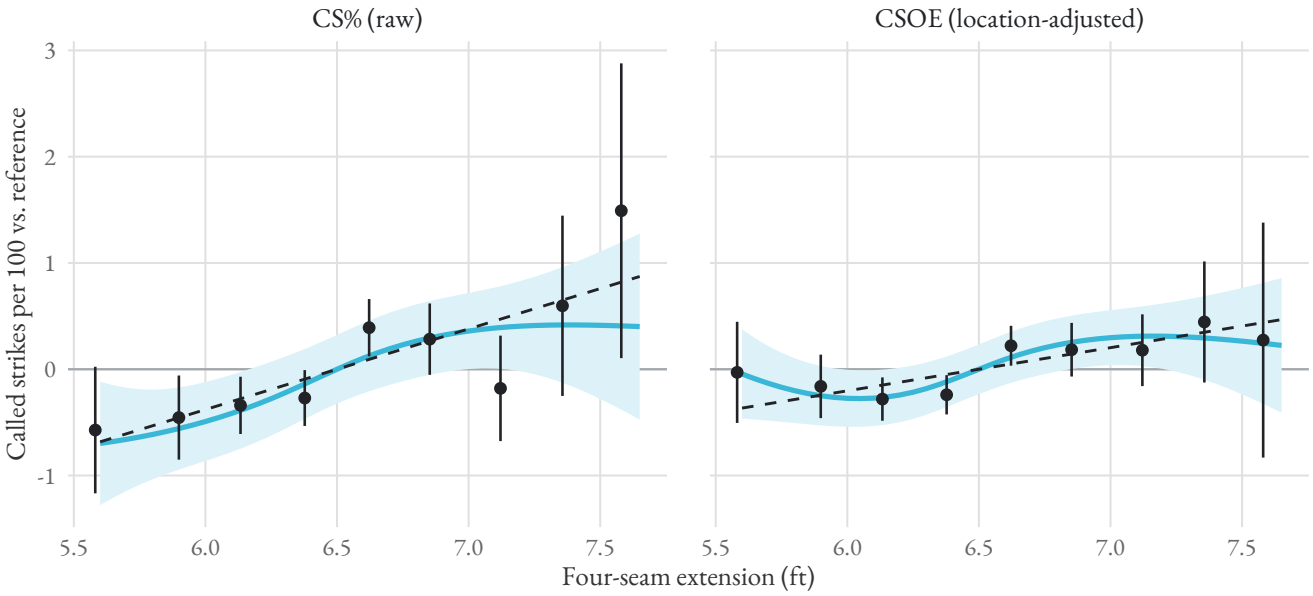


Figure 3: Four-seamers: called strikes per 100 pitches relative to a 6.5 ft extender with the same velocity, movement, release point and arm angle. Points are extension buckets (95% intervals). Dashed line is the straight per-foot fit; the cyan band is a flexible curve with its 95% interval. Height is not held constant here.

may come from extending far *for your size*. Listed heights for all 1,246 pitchers came from the MLB StatsAPI.

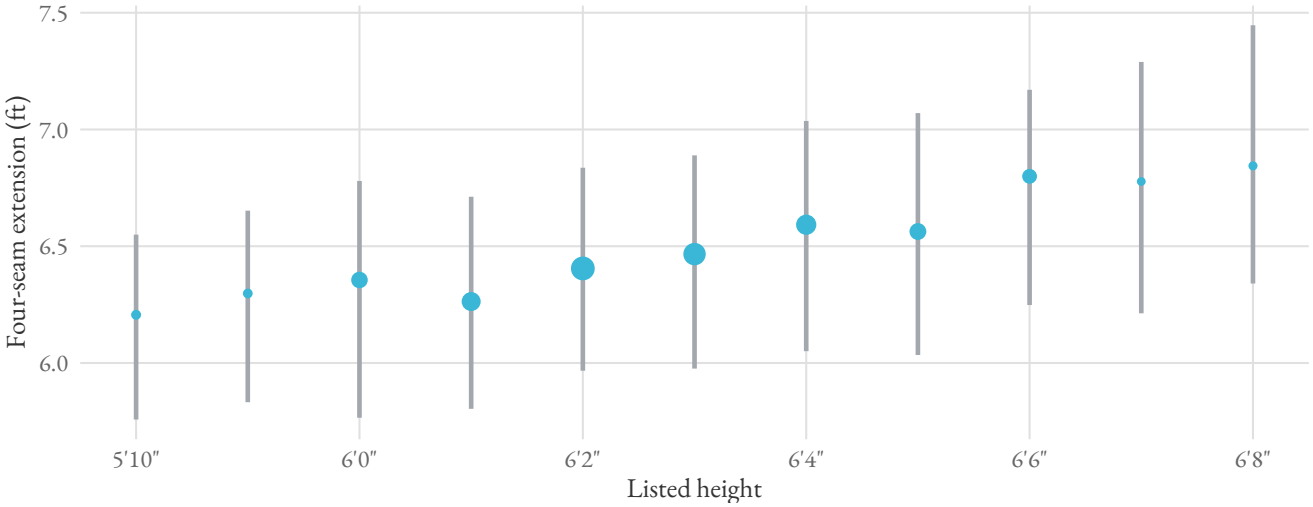


Figure 4: Four-seam extension by listed height. Points are the average; bars run from the 10th to the 90th percentile of pitcher-seasons. Heights with fewer than 10 pitcher-seasons are left out.

Taller pitchers extend farther, but only by about 0.9 ft per foot of height. Height explains just 14% of the spread: pitchers of the same height differ by about 0.38 ft (one standard deviation). So most of extension is delivery, not size. "Extension for size" below is a pitcher's four-seam extension minus what is typical for his height.

Table 1: Four-seamers, called strikes per 100 pitches: per foot of extension, with and without height in the model.

Outcome	Model	Per ft	Velo benchmark	Extra	p	Per inch of height
CS%	Raw extension	+0.76	−0.73	+1.49	<0.0001	
CS%	Extension for size + height	+1.09	−0.73	+1.82	<0.0001	−0.05 (se 0.06)
CSOE	Raw extension	+0.41	−0.50	+0.91	<0.0001	
CSOE	Extension for size + height	+0.35	−0.50	+0.85	0.00011	+0.05 (se 0.04)

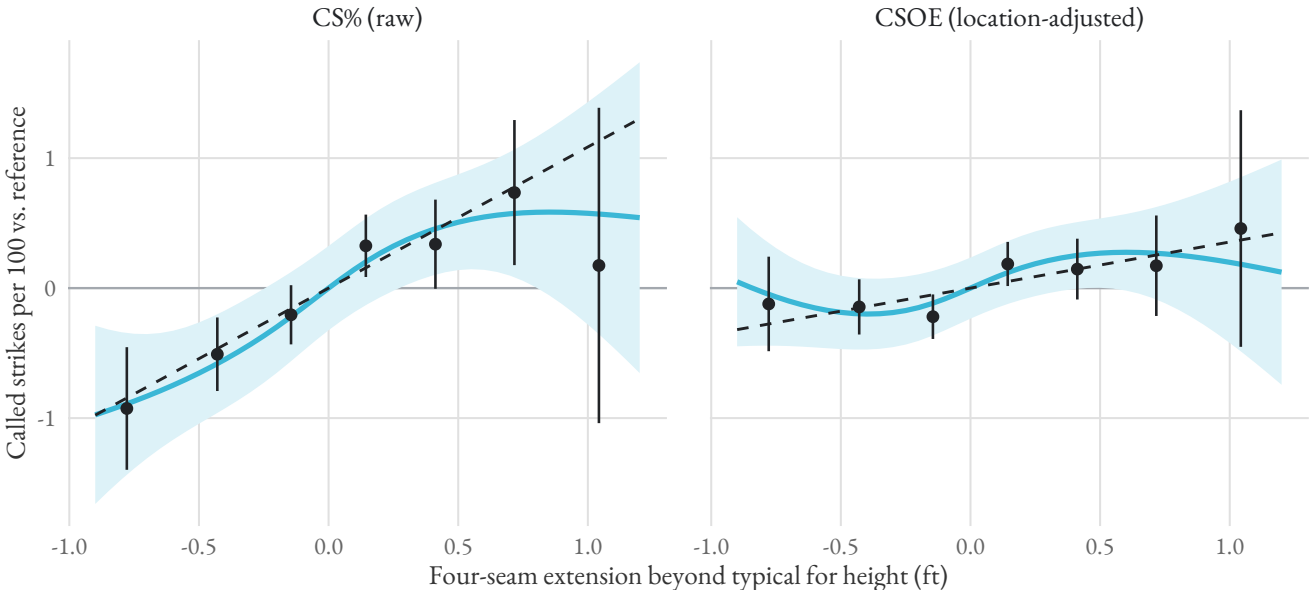


Figure 5: Four-seamers, height held constant: called strikes per 100 pitches by extension for size (ft beyond what is typical for the pitcher's height), relative to a pitcher at the typical extension for his height. Same layout as Figure 3.

- **Holding height constant doesn't shrink the effect.** The raw called strike gain grows to +1.09 per foot, because taller pitchers get slightly fewer called strikes at the same extension (−0.05 per inch, not significant). The location-adjusted gain barely moves (+0.35).
- **There is no "big for his size" bonus.** Pitchers more than 0.45 ft beyond the norm for their height (about the top 10%) show no extra slope (−0.76 per foot for CS%, p 0.6; −0.05 for CSOE, p 0.97). The curve doesn't bend (p 0.4 and 0.11).
- **Extension as a share of height adds nothing.** Once raw extension and height are in the model, adding extension as a percentage of height changes nothing (p = 0.66 for CS%, 0.82 for CSOE).

> **Takeaway.** What matters is how close to the plate the ball comes out, not how far that is for the pitcher's body. Measured against height, the biggest extenders still get no bonus.

PART 5

Other pitches and the whole arsenal

- **Four-seamers and changeups/splitters gain.** Changeups and splitters pick up +0.93 called strikes above expected per foot. They are thrown to look like the fastball, so the same timing edge plausibly carries over.
- **Sinkers show nothing clear** (+0.12, wide interval).

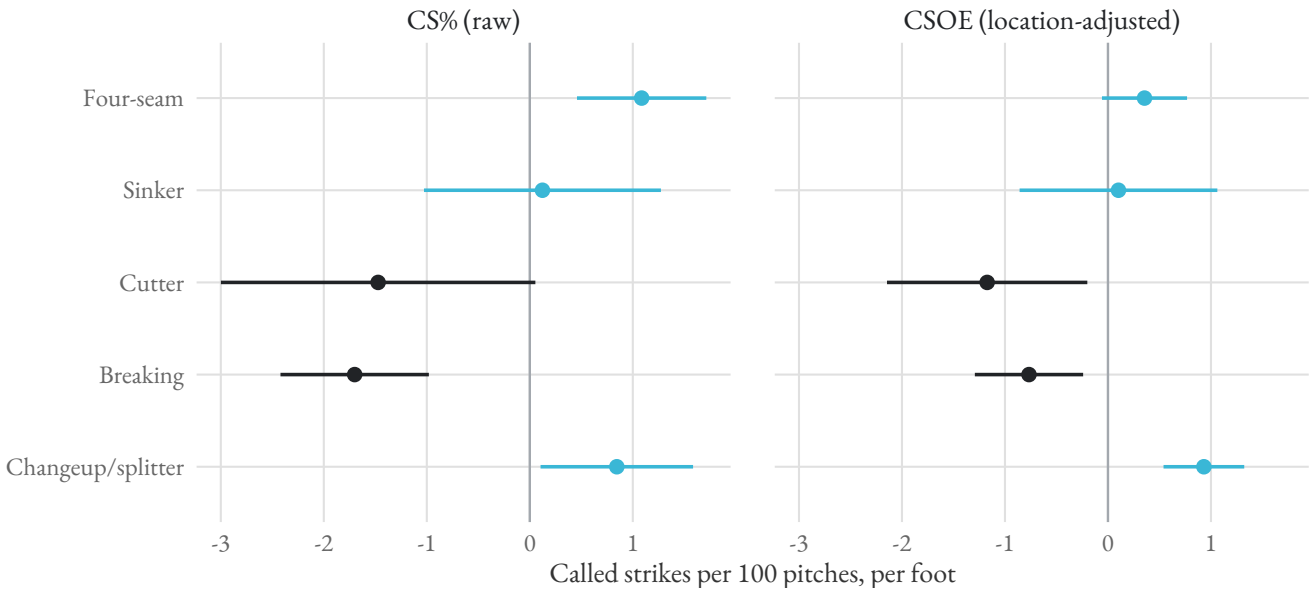


Figure 6: Called strikes per 100 pitches from one foot of extension for size, by pitch type, height held constant. Bars are 95% intervals.

- **Cutters and breaking balls lose.** Breaking balls give up −1.70 called strikes per foot (−0.77 location-adjusted), and cutters −1.47. One possible reason: extension shortens the hitter's look at a pitch, and on pitches that start out of the zone and break in, that may make him swing more rather than take.
- **Across the whole arsenal it nets out.** Keyed on four-seam extension and holding fastball share constant, a pitcher's overall called strike rate moves −0.03 per 100 pitches per foot (+0.13 location-adjusted). Neither is distinguishable from zero.

> **Takeaway.** The extension edge is real on four-seamers and changeups, but breaking balls and cutters give it back. A long extender doesn't steal more called strikes overall; he steals them on his fastball.

APPENDIX

Data, methods and checks

Data

MLB regular-season pitches, 2021-2026, from the Baseball Savant files on the drive (`MLB Data/<year>/MLB`). Left out: automatic balls and strikes, pitchouts, hit-by-pitches, bunt attempts, position-player pitching, knuckleballs and eephus pitches, and extension outside 4-8.5 ft.

Table 2: Pitches and modeling groups by season (groups need 150+ pitches).

Season	Pitches	CS%	Pitcher-season-pitch groups	Pitchers
2021	703,247	16.5%	1,461	589
2022	701,719	16.4%	1,475	563
2023	711,293	16.5%	1,514	554
2024	702,251	16.5%	1,491	531
2025	703,865	16.4%	1,511	547
2026	707,199	16.3%	1,549	538

Methods

- **Expected called strikes (xCS).** A gradient-boosted model predicts each pitch's called-strike chance from location only: plate side, height scaled to the batter's zone, count, batter side, same-side matchup, pitch group (fastball, breaking, offspeed) and season. It knows nothing about the pitcher. It is cross-fit by game, so no pitch is scored by a model that saw it (log-loss 0.296 vs. 0.447 for a flat rate). CSOE = called strikes minus xCS.
- **Unit.** Pitcher-season-pitch type with 150+ pitches. Weighted least squares, weighted by pitches, with standard errors clustered by pitcher.
- **Controls.** Velocity, IVB, HB (arm side positive), release height, release side, arm angle, season, and pitch type where groups combine several.
- **Perceived-velocity benchmark.** The slope of Statcast's effective speed minus release speed on extension, for four-seamers: 1.72 mph per foot. "Extra beyond perceived velocity" = the extension slope minus 1.72 times the velocity slope, tested with a Wald test.
- **Tail tests.** An extra slope above 7.0 ft of raw extension, or above +0.45 ft of extension for size (each about the top 10%), plus a natural cubic spline (4 df) compared with the straight line by an F-test.
- **Height.** Listed heights from the MLB StatsAPI people endpoint. These are current listings, not by season, and listed heights are often rounded or generous.

Table 3: All pitch types: called strikes per 100 pitches per foot of extension. The top-10% slope is the extra slope above 7.0 ft (raw) or +0.45 ft (for size).

Pitch	Outcome	Model	Per ft (se)	Velo benchmark	Extra	p (extra)	Top-10% slope (p)
Four-seam	CS%	Raw	+0.76 (0.26)	−0.73	+1.49	<0.0001	−0.24 (0.87)
Four-seam	CS%	For size + height	+1.09 (0.32)	−0.73	+1.82	<0.0001	−0.76 (0.6)
Four-seam	CSOE	Raw	+0.41 (0.19)	−0.50	+0.91	<0.0001	−0.09 (0.94)
Four-seam	CSOE	For size + height	+0.35 (0.21)	−0.50	+0.85	0.00011	−0.05 (0.97)
Sinker	CS%	Raw	+0.16 (0.49)	−0.75	+0.91	0.078	+0.24 (0.94)
Sinker	CS%	For size + height	+0.12 (0.59)	−0.75	+0.87	0.15	+1.67 (0.64)
Sinker	CSOE	Raw	+0.38 (0.40)	−0.41	+0.79	0.066	+0.22 (0.92)
Sinker	CSOE	For size + height	+0.10 (0.49)	−0.40	+0.50	0.33	+2.66 (0.22)
Cutter	CS%	Raw	−1.41 (0.57)	+0.09	−1.50	0.0064	−2.38 (0.54)
Cutter	CS%	For size + height	−1.47 (0.78)	+0.09	−1.56	0.04	+0.71 (0.89)
Cutter	CSOE	Raw	−1.02 (0.34)	+0.06	−1.08	0.00081	+0.54 (0.8)
Cutter	CSOE	For size + height	−1.17 (0.50)	+0.06	−1.23	0.0098	+3.33 (0.33)
Breaking	CS%	Raw	−1.41 (0.30)	−0.04	−1.36	<0.0001	+4.62 (0.014)
Breaking	CS%	For size + height	−1.70 (0.37)	−0.03	−1.67	<0.0001	+3.85 (0.068)
Breaking	CSOE	Raw	−0.73 (0.21)	+0.21	−0.93	<0.0001	+0.92 (0.48)
Breaking	CSOE	For size + height	−0.77 (0.27)	+0.21	−0.97	0.00034	+1.43 (0.35)
Changeup/splitter	CS%	Raw	+1.05 (0.29)	−0.10	+1.15	0.00021	−0.13 (0.96)
Changeup/splitter	CS%	For size + height	+0.84 (0.38)	−0.10	+0.94	0.017	+2.63 (0.25)
Changeup/splitter	CSOE	Raw	+0.82 (0.16)	−0.12	+0.93	<0.0001	+0.66 (0.54)
Changeup/splitter	CSOE	For size + height	+0.93 (0.20)	−0.11	+1.05	<0.0001	+1.14 (0.22)

Checks

Table 4: Robustness. Same pitcher, season to season: four-seamers with pitcher fixed effects (pitchers with 2+ seasons, raw extension). Whole arsenal: every pitch a pitcher threw in a season (500+), keyed on four-seam extension for size, with height and four-seam share held constant.

Check	Outcome	Groups	Per ft (se)	Velo benchmark	Extra	p
Same pitcher, season to season	CS%	2,186	+2.02 (0.81)	−0.62	+2.65	0.002
Same pitcher, season to season	CSOE	2,186	+0.85 (0.70)	−0.17	+1.02	0.17
Whole arsenal (pitcher-season)	CS%	2,226	−0.03 (0.18)	−0.18	+0.15	0.44
Whole arsenal (pitcher-season)	CSOE	2,226	+0.13 (0.17)	−0.07	+0.21	0.25

- **Same pitcher, season to season.** When a pitcher's own four-seam extension moves from one season to the next, his raw called strike rate follows (+2.02 per foot, p 0.002 against the benchmark). That rules out a fixed trait of long extenders, such as body type, as the whole explanation. The location-adjusted version points the same way but isn't significant (p 0.17).
- **Multiple tests.** The report runs about 20 extension tests. The four-seam findings survive any reasonable correction ($p < 0.0001$). One raw-CS% tail slope for breaking balls ($p = 0.014$, raw extension) disappears once location is controlled, so it is treated as noise.

Limits

- No adjustment for umpire or catcher framing. Both would need to line up with pitcher extension to bias the results, which seems unlikely.
- 2026 has the ABS challenge system. Overturned calls show up as final calls, and season effects absorb any league-wide shift.
- Statcast extension is measured, not perceived. How it plays to hitters probably differs from pitcher to pitcher (for example, hiding the ball), and this study can't see that.